

Healthcare Workforce Analytics: Nursing Home Staffing, Quality, and Compliance Analysis

Introduction

Healthcare workforce availability plays a critical role in the quality of care delivered within nursing homes. Adequate staffing levels, particularly among registered nurses (RNs), licensed practical nurses (LPNs), and certified nursing assistants (CNAs), are essential for maintaining resident safety, meeting regulatory requirements, and achieving positive health outcomes. As workforce shortages continue to affect healthcare organizations nationwide, understanding the relationship between staffing practices and facility performance has become increasingly important.

This project analyzes payroll-based staffing data, facility quality ratings, health inspection results, and compliance measures from nursing homes across the United States. Using PostgreSQL for data analysis and Power BI for visualization, the project examines how staffing levels, nurse workforce availability, contract labor utilization, facility performance, and regulatory compliance influence overall nursing home quality ratings.

The analysis focuses on four key areas: executive workforce trends, facility performance, staffing breakdowns, and compliance and penalties.

Executive Summary

The Executive Summary dashboard provides a high-level overview of workforce trends and nursing home performance across the United States. Several key findings emerged from the analysis.

First, staffing levels were positively associated with facility quality ratings. Facilities receiving higher CMS Overall Ratings consistently reported greater staff hours per resident per day than lower-rated facilities. This suggests that adequate staffing remains a significant contributor to quality outcomes.

Second, RN staffing demonstrated a particularly strong relationship with quality performance. As facility ratings increased, average RN hours per resident also increased. Facilities with stronger RN coverage generally achieved higher quality scores, indicating the importance of skilled nursing oversight in long-term care settings.

Third, ownership structure appeared to influence performance outcomes. Government-operated and nonprofit facilities generally achieved higher average quality ratings than many for-profit organizations. Differences in staffing levels and workforce investment may contribute to these performance variations.

Finally, facilities with lower quality ratings tended to exhibit greater reliance on contract staffing. While contract nurses can help address workforce shortages, heavy dependence on temporary staffing did not appear to correspond with improved quality outcomes.

Overall, the executive analysis suggests that workforce investment, particularly in permanent nursing staff, is closely linked to nursing home quality performance.

Facility Performance Analysis

The Facility Performance Analysis dashboard examined how individual facilities performed across quality measures and operational indicators.

Analysis of facility ratings revealed substantial variation among nursing homes nationwide. While many facilities achieved average ratings, only a smaller percentage attained the highest quality classifications. Top-performing facilities consistently demonstrated stronger staffing levels and better survey performance.

Health survey scores showed a clear relationship with overall facility ratings. Facilities receiving higher ratings generally reported stronger inspection results and fewer regulatory deficiencies. This finding

supports the CMS rating methodology, which places significant emphasis on health inspections and resident care outcomes.

The analysis also examined financial penalties and enforcement actions. Facilities with lower ratings tended to incur higher fines and regulatory penalties, suggesting that workforce challenges and compliance issues often occur simultaneously. In contrast, higher-performing facilities generally experienced fewer enforcement actions and maintained stronger compliance records.

These findings indicate that facility performance is influenced by multiple interconnected factors, including staffing practices, regulatory compliance, and quality management processes.

Staffing Breakdown Analysis

The Staffing Breakdown Analysis dashboard focused on the composition of the nursing home workforce and the contribution of various healthcare professionals.

The analysis examined staffing hours reported for registered nurses (RNs), licensed practical nurses (LPNs), certified nursing assistants (CNAs), and contract nursing staff. CNA staffing represented the largest portion of direct resident care hours across most facilities, reflecting the critical role CNAs play in supporting daily resident needs.

RN staffing emerged as one of the strongest predictors of facility quality. Facilities with higher RN hours per resident consistently achieved stronger overall ratings and better health survey outcomes. Because RNs are responsible for clinical oversight, care coordination, and complex medical decision-making, their presence appears to contribute significantly to organizational performance.

The analysis also revealed substantial variation in contract staffing utilization. Some facilities relied heavily on agency nurses and temporary staffing arrangements, while others maintained predominantly permanent workforces. Facilities with greater contract staffing utilization generally demonstrated lower quality ratings, suggesting that workforce stability may be an important factor in achieving consistent care outcomes.

Overall, the staffing breakdown highlights the importance of maintaining adequate nursing coverage and investing in permanent healthcare personnel to support resident care quality.

Compliance and Penalties Analysis

The Compliance and Penalties Analysis dashboard evaluated regulatory performance across nursing homes using health inspection scores, deficiency measures, fines, and enforcement actions.

Facilities with lower CMS ratings generally experienced poorer health survey outcomes and accumulated more regulatory deficiencies. These organizations were also more likely to receive financial penalties and enforcement actions from regulatory agencies.

Conversely, higher-rated facilities demonstrated stronger compliance performance. These facilities reported fewer deficiencies, lower penalty amounts, and better overall inspection results. The findings suggest that compliance outcomes are closely connected to workforce effectiveness and operational management practices.

Financial penalties varied considerably among facilities, with some organizations accumulating significant fine amounts due to repeated deficiencies or serious regulatory violations. These enforcement actions often coincided with lower staffing levels and weaker quality ratings.

The compliance analysis demonstrates that maintaining appropriate staffing levels is not only important for resident care but also for reducing regulatory risk and improving organizational performance.

Conclusion

This project explored the relationship between workforce staffing, facility performance, and regulatory compliance within U.S. nursing homes. Analysis of staffing data, quality ratings, inspection outcomes, and enforcement measures revealed several important findings.

Facilities with higher staffing levels, particularly RN staffing, consistently achieved stronger quality ratings and better inspection results. Government and nonprofit facilities generally outperformed many for-profit organizations, while facilities with heavy reliance on contract staffing often experienced lower quality outcomes. Additionally, lower-rated facilities were more likely to incur fines, deficiencies, and regulatory penalties.

These findings reinforce the importance of workforce investment in long-term care settings. Maintaining adequate staffing levels, supporting workforce stability, and prioritizing skilled nursing coverage may contribute to improved resident outcomes, stronger regulatory compliance, and higher overall facility performance.

By combining SQL-based data analysis with interactive Power BI visualizations, this project demonstrates how healthcare workforce analytics can be used to identify operational trends, evaluate quality performance, and support evidence-based decision-making within healthcare organizations.